

Reconstruction Without Reallocation: How Did Chilean Manufacturing Capital Adjust After the 2010 Maule Earthquake Shock?

Sitong Liu*

August 27, 2026

Abstract

In February 2010, Chile was struck by the Maule earthquake, which measured 8.8 on the moment magnitude scale and dealt a severe blow to the local manufacturing sector. This paper treats the Maule earthquake as an exogenous shock to Chile's manufacturing sector to examine how the sector recovered in the aftermath of the earthquake, whether this recovery spilled over into the labor market, and whether the recovery had heterogeneous effects on unskilled and skilled workers. I address this question using a difference-in-differences design across industries and regions. The results show that, among the variables examined—output, employment, labor productivity, the labor share of output, proxy measures of profits, and investment—only investment responded to capital reconstruction, while no significant changes were observed in the other indicators. However, since the severely affected central urban regions already had higher investment levels than other areas prior to the disaster, I interpret this evidence as suggestive rather than conclusive. The paper primarily demonstrates that reconstruction restored the capital stock but did not alter the scale of production or the allocation of resources.

Keywords: natural disasters, capital-skill complementarity, labor markets, Chile.

JEL: J23, J31, Q54, O14, R11.

*Undergraduate Honors Program in Economics, New York University. This is an independent research project. I thank ... All errors are my own.

1 Introduction

In the early hours of February 27, 2010, the 8.8-magnitude Maule earthquake struck Chile. It was the sixth-largest earthquake in recorded history and the second strongest in Chile’s history. More than 100 aftershocks with magnitudes of 5.0 or higher followed. Over 1.8 million people were affected, infrastructure across the country suffered severe damage, and economic losses reached \$30 billion, accounting for approximately 18% of Chile’s GDP. The three hardest-hit regions, Maule, Bío-Bío, and O’Higgins, constitute the heartland of Chile’s manufacturing sector. Together, these regions account for more than a quarter of Chile’s manufacturing output, spanning industries such as food processing, beverages, steel, paper, shipbuilding, and oil refining, while also contributing approximately half of the country’s agricultural and forestry output (Bárcena et al., 2010). According to a study using synthetic control methods, economic activity in the Maule Region declined by approximately 5% compared to a counterfactual scenario in the three years following the earthquake (Díaz et al., 2024). Figure 1 shows the distribution of seismic intensity.

As can be seen, this earthquake was plausibly exogenous to the industrial and labor markets. It had a massive impact, and the intensity of ground shaking it generated attenuated with distance from the epicenter and varied with local soil conditions (Wald et al., 1999; Wald & Allen, 2007). Therefore, after controlling for fixed effects, the intensity of shaking is plausibly uncorrelated with a region’s pre-existing manufacturing and labor-market trends. I can thus attempt to answer the following question: How does a labor market adjust when an economy suddenly loses a large amount of physical capital overnight due to an exogenous disaster? Krusell et al. (2000) proposed the “capital-skill complementarity (CSC)” hypothesis to explain the relationship between equipment capital and the labor market: equipment capital and skilled labor are complementary in production. I can thus propose the initial hypothesis that, after an earthquake destroys most of the local physical capital, the local economy will require more skilled workers to rebuild capital, thereby driving up wages for skilled workers.

To test whether this theory applies to this earthquake disaster, I aggregate establishment-level data into industry $\times$ region $\times$ year cells and construct a difference-in-differences specification with an interaction term between the time indicator variable (post-2010) and continuous regional MMI values. The industry $\times$ region fixed effects capture the time-invariant characteristics of each cell, including MMI levels. The industry $\times$ year fixed effects capture any nationwide industry-specific shocks, such as changes in copper prices, global demand for processed foods, and the national business cycle. Thus, I compare establishments across regions within the same industry and year that were affected by the earthquake to varying

degrees.

Reconstruction unfolds over several years rather than instantaneously. The immediate aftermath is dominated by relief, damage assessment, and financing rather than rebuilding, so the investment response should appear with a delay, peaking in 2014–2015 rather than at the moment of the quake. Consistent with a slow recovery, [Díaz et al. \(2024\)](#) find that Chilean per-capita GDP remained roughly 5% below its counterfactual for years after the earthquake. This motivates the central question of the paper: during reconstruction, how did the manufacturing labor market adjust, through capital, through workers, or through wages?

The paper makes three points in response to the above questions.

(1) A reconstruction-investment pulse. Areas that were more severely affected by the earthquake saw an increase in total investment following the disaster, peaking in 2014 before declining. For every one-unit increase in a region’s MMI, the post-earthquake peak investment in that region rose by approximately 0.58 log points. This estimate withstood a wild cluster bootstrap and a series of robustness tests. However, it was found to be sensitive to the faster pre-earthquake investment growth trend in affected regions.

(2) Reconstruction occurred after the earthquake, but the allocation of resources did not change. Investment was the only variable to show a significant change post-earthquake. I did not observe significant changes in other variables such as output, employment, labor productivity, the labor share of output, or proxy indicators of corporate profits (see Table 5). Therefore, this suggests that reconstruction merely replaced capital destroyed by the earthquake rather than adding new productive capacity, and thus did not alter the allocation structure between capital and labor. This conclusion is consistent with the fact that Chile’s value added remained below its counterfactual path several years later ([Díaz et al., 2024](#)). This is the key finding of the paper and the fundamental reason why the effects of post-earthquake reconstruction did not spill over into the labor market.

(3) The expected “capital-skill complementarity” theory has not been confirmed. If reconstructed capital and skilled labor were complementary, the post-disaster skill premium should have risen. However, this pattern did not materialize following the Chilean earthquake. Yet my aim is not to refute this theory, as the available data and the reduced-form regression I employ do not solely examine the impact of equipment on the skill premium. They also account for the effects of variables such as structures, labor, and output—all of which were similarly affected by the earthquake—on the skill premium. Furthermore, the wage evidence is not sufficiently robust. I find a weak and imprecise upward trend in unskilled wages following the disaster. A candidate mechanism is that local reconstruction of destroyed buildings and infrastructure after the disaster led to increased

employment in the construction sector. I therefore used household data to test this hypothesis but found no significant results.

Pre-trends and the identification strategy. When I regressed the various outcomes on “post-earthquake $\times$ regional intensity,” I found that the hardest-hit areas were precisely Chile’s industrial heartland (Santiago, Maule, and Bío-Bío), which had already been experiencing faster investment growth than outlying regions prior to the earthquake. To address this issue, I recovered the ENIA files with establishment identifiers from before 2008 and extended the pre-earthquake period back to 2003 to capture regional trends. I then incorporated region-specific trends estimated solely within the pre-earthquake window. Even so, this study cannot completely disentangle pre-earthquake investment growth from post-earthquake investment effects, so the conclusions presented here are tentative.

Related literature. In the economics of natural disasters, a significant body of research focuses on regional output following major shocks, as well as how populations adapt and recover (Aksoy et al., 2024; Díaz et al., 2024; Benguria, 2022), while a smaller body of research examines labor market conditions in affected regions. The study most closely aligned with this paper is Belasen & Polachek (2009), who found that hurricanes in Florida increased local income but reduced employment. Building on these studies, this paper also examines how manufacturing establishments adjust their total investment, asset mix, and occupational composition following a disaster.

Second, some studies have used capital cost shocks to investigate the extent to which such shocks are passed on to workers (Suárez Serrato & Zidar, 2016; Fuest et al., 2018), while Notowidigdo (2020) examined how changes in local labor demand affect skill groups differently. I adopt a similar logic, but with fundamental differences from previous studies, which also lead to different results. The shocks in this paper involve physical destruction, such as damage to factory machinery, buildings, and infrastructure, rather than changes in tax rates or the cost of capital. Furthermore, I find that the physical destruction caused by earthquakes is rarely passed on to workers. Even when it is, the pass-through is not significant.

Third, this paper provides an empirical case study of “capital-skill complementarity.” The capital-skill complementarity theory proposed by Krusell et al. (2000) emphasizes that capital equipment increases the relative demand for skilled labor. Meanwhile, Lewis (2011) argues that the local labor supply, in turn, shapes an establishment’s choice of equipment. The Chilean earthquake provides an empirical case: in the aftermath of the disaster, local manufacturers increased investment in equipment to restore the economy, yet the skill premium did not rise. However, this conclusion is based on the results of a reduced-form regression and is not entirely robust. Therefore, it is insufficient to refute the validity of

Krusell’s theory in this specific case.

Finally, this paper draws on the pre-trends approach of [Roth \(2022\)](#) to address the issue that the hardest-hit regions already experienced faster investment growth prior to the disaster. Section 2 presents the concepts and theories discussed in this paper. Section 3 introduces the data. Section 4 outlines the empirical strategy. Sections 5–6 present the results and mechanisms. Sections 7–8 demonstrate the exclusion of the out-migration channel and the treatment of parallel trends. Section 9 discusses the effects and external validity. Section 10 concludes the paper.

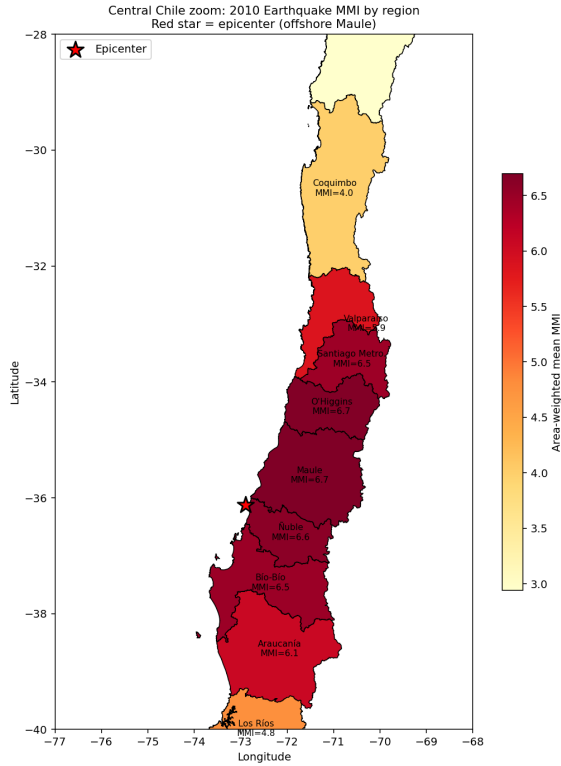


Figure 1: Ground-shaking intensity (MMI) of the 2010 Maule earthquake by region. The red star marks the offshore epicenter. Central regions were severely shaken. The far north and south were essentially unaffected.

2 Conceptual Framework

According to the capital-skill complementarity theory proposed by [Krusell et al. \(2000\)](#), let k_s denote structures (buildings used for production), k_e denote equipment, s denote skilled labor, and u denote unskilled labor. The complementarity between equipment and skilled

labor is stronger than that between equipment and unskilled labor:

$$Y = A k_s^\alpha \left[\mu u^\sigma + (1 - \mu) (\lambda k_e^\rho + (1 - \lambda) s^\rho)^{\sigma/\rho} \right]^{(1-\alpha)/\sigma}. \quad (1)$$

ρ measures the ease of substitution between equipment and skilled labor; the smaller it is, the harder it is for the two to substitute for one another. σ measures the ease of substitution between equipment and unskilled labor; the larger it is, the easier it is for the two to substitute for one another. If $\sigma > \rho$, then equipment is harder to substitute for skilled labor than for unskilled labor. In other words, machines require highly skilled workers to operate them, but are less selective when it comes to unskilled workers.

We can therefore conclude that the relative marginal product of skilled labor rises as the stock of equipment increases; consequently, the skill premium $\omega = w_s/w_u$ will rise as the amount of equipment allocated to skilled workers increases:

$$\frac{\partial \ln \omega}{\partial \ln k_e} > 0 \quad \text{iff} \quad \sigma > \rho. \quad (2)$$

Equation (2) provides a standard prediction: if equipment stock increases, the wage premium of skilled labor relative to unskilled labor will increase accordingly. Since plant capital is treated as a separable Cobb-Douglas factor within Equation (1), it does not carry this implication, which is consistent with the “capital-specificity” noted by Krusell et al.

This baseline formula overlooks a key feature of post-disaster reconstruction. Reconstruction itself constitutes a form of production activity, involving labor-intensive manual and construction work, such as demolition, site clearance, and the installation and construction of new facilities and equipment, which relies heavily on unskilled workers. Therefore, when severely affected regions rely more heavily on post-disaster facility reconstruction, they have a greater need for unskilled workers, leading to an outward shift in the demand for unskilled labor, where $\Delta \ln u^d > 0$. However, the supply of unskilled labor is not infinitely elastic; this rise in demand will push w_u upward, causing the skill premium we obtain to be underestimated, because of the opposite of the direction of effect in Equation (2). Consequently, the net effect on ω is theoretically uncertain:

$$\underbrace{\frac{\partial \ln \omega}{\partial \ln k_e}}_{>0 \text{ (CSC)}} \quad \text{vs.} \quad \underbrace{\frac{\partial \ln \omega}{\partial \ln u^d}}_{<0 \text{ (reconstruction demand)}}. \quad (3)$$

This framework raises two empirical questions that this paper seeks to address: H1 (Investment): Does the shock increase investment in the affected region? H2 (Allocation):

Where does this investment manifest itself—in output, employment, factor allocation, or skill premiums? Equation (2) provides an expectation that this investment will ultimately manifest itself in rising skill premiums. However, Equation (3) casts doubt on this expectation. Because the shock simultaneously alters k_e , k_s , u^d , and output, the reduced-form regression performed in this paper captures the net effect of k_e , k_s , u^d , and output as a whole, rather than a pure, isolated $\partial \ln \omega / \partial \ln k_e$. The data confirm H1, but the result for H2 is that the investment is not reflected in the actual marginal rate of return nor in factor allocation. Therefore, I argue that post-earthquake reconstruction was limited to restoring destroyed capital and infrastructure, without altering the total volume of production or the pattern of allocation between labor and capital. The prediction of a skill premium was not confirmed, nor was it supported by existing data for a clean test.

3 Data

3.1 Manufacturing microdata

The Annual National Industrial Survey (*Encuesta Nacional Industrial Anual*, ENIA) is conducted annually by the Chilean National Institute of Statistics (INE). The survey covers manufacturing establishments with ten or more employees. I use survey data from 2003 to 2017, excluding 2012 because the response rate from establishments was unusually low that year. Each survey includes data reported by establishments on employment, wages and salaries by occupational category, gross production value (*valor bruto de la producción*, VBP), intermediate consumption, taxes, and fixed-asset acquisitions and book values by asset category. Appendix A details the data processing procedures.

Three points warrant emphasis. First, public-use files from 2008 onward lack stable establishment identifiers, so establishment-level data is unavailable; I can only analyze industry-region-year cells. Additionally, the files I reconstructed for the period prior to 2008 contain stable identifiers (NUI), which can be used to extend the data to the pre-earthquake period. Second, ENIA does not directly survey capital stock; therefore, the data do not directly reflect how much capital was destroyed by the earthquake or how much was replenished. Consequently, I directly use annual acquisition expenditures as an indicator of capital formation. I categorize these expenditures into structures (land and buildings), machinery, and a broad equipment measure (machinery, vehicles, and software). Third, starting in 2013, the industrial classification standard used in the statistics changed from ISIC Rev. 3 to Rev. 4. I use the official UN concordance table to reconcile the coding differences between the two versions, prioritizing the UN standard and classifying industries with discrepancies based on

their weights.

3.2 Skill, wages, and the limits of the employment data

To test the theory of “capital-skill complementarity,” I must first divide workers into two categories: skilled workers and unskilled workers. ENIA reports employment figures and wages by occupational category, so in theory, I could calculate the proportion of skilled workers based on the number of workers in each category across regions. However, ENIA’s classification of these two categories is somewhat vague and does not provide a clear distinction between unskilled and skilled workers. Fortunately, the wage category is broken down into monthly wages of technical/professional workers and monthly wages of unskilled workers, so I use these two wage series to calculate the skill premium. The skill premium is defined as the log of the ratio of the monthly wages of technical/professional workers to the monthly wages of unskilled workers.

3.3 Earthquake intensity

I use the Modified Mercalli Intensity (MMI) calculated by the USGS ShakeMap ([U.S. Geological Survey, 2010](#)) to measure earthquake intensity. Unlike the magnitude of 8.8, which represents the total energy of the entire earthquake, the MMI indicates the actual intensity of shaking felt at a specific location. This allows me to distinguish between severely affected and lightly affected areas based on the intensity of the shaking. By overlaying the intensity contours from ShakeMap onto administrative district maps and then applying area-weighted averaging and an equal-area projection, I can calculate the average intensity for each administrative district. The average MMI across regions ranges from approximately 2.9 (in the northern desert regions) to 6.7 (in Maule and O’Higgins). Areas that were virtually unaffected by the earthquake are assigned the lower-limit intensity of 2.0 and serve as the control group. To ensure consistency in regional coding across the data, I use the classification standard of the 15 major regions. Therefore, the Ñuble region is still grouped under the Bío-Bío region. I treat continuous intensity as the main explanatory (treatment) variable and use a binary indicator variable for high intensity ($\text{MMI} \geq 6$) as a robustness test.

3.4 Sample and descriptive statistics

I aggregate the data into industry $\times$ region $\times$ year cells, calculate the weighted sum of aggregate economic indicators such as investment and employment, and compute the weighted average of wages. I select food products (ISIC 10), wood products (ISIC 16), and basic met-

als (ISIC 24) as the sample industries because these three sectors are widely distributed and have comparable levels of development across both low- and high-intensity regions, ensuring that the comparison between high- and low-intensity regions is not confounded by differences in industry composition. I report the results of the robustness check using the entire manufacturing sample. Table 1 presents the conditions in severely and lightly affected areas prior to the earthquake. The results indicate that the severely affected areas were substantially ahead of the lightly affected areas in terms of investment, employment, and output. This suggests that the most severely affected central regions were already the industrial heartland of Chile. Therefore, I suspect that the two groups of regions were likely already following different growth trajectories prior to the earthquake. This is why I estimate pre-earthquake trends and project them forward in Section 4, and test their robustness in Section 8.

Table 1: Summary statistics, pre-earthquake cells (2003–2009)

	High MMI (≥ 6)	Low MMI (< 6)
Mean MMI	6.488	4.012
Investment (CLP, cell sum)	32,690,854	24,311,385
Employment (cell sum)	24,435	8,823
Gross output VBP (cell sum)	1,471,919,965	1,122,781,240
Log skill premium	-0.708	-0.559
Cell-years	83	128

Notes. Weighted means across industry $\times$ region $\times$ year cells (Food, Wood, Basic metals) over 2003–2009. High-MMI regions are those with area-weighted intensity ≥ 6 .

4 Empirical Strategy

4.1 Specification

The baseline difference-in-differences specification is

$$y_{jrt} = \beta (\text{Post}_t \times \text{MMI}_r) + \gamma_{jr} + \delta_{jt} + \varepsilon_{jrt}, \quad (4)$$

where j indexes 2-digit industry, r region, and t year; $\text{Post}_t = \mathbf{1}[t \geq 2010]$; MMI_r is the region's area-weighted intensity; γ_{jr} are industry $\times$ region fixed effects; and δ_{jt} are industry $\times$ year fixed effects. The industry $\times$ region effects absorb the main effect of MMI and any time-invariant cell heterogeneity; the industry $\times$ year effects absorb any shock common to an industry nationwide each year. β is the coefficient of interest: it captures the differential

change in the outcome after 2010 associated with a one-unit-higher MMI in a region—that is, how much more the outcome moved after the earthquake in regions shaken one unit harder. I replace MMI_r with a binary variable (indicating whether an area is severely affected) as a robustness check, and weight the cells by size, with standard errors clustered by region.

4.2 Identification, pre-trends, and detrending

However, Equation (4) is constrained by the parallel-trends assumption of the difference-in-differences method, which holds that if the earthquake had never occurred, development in regions with high MMI and low MMI would have followed parallel trends within each industry. Yet, with 2009 as the base year and 2003–2009 as the pre-earthquake phase, this parallel-trends assumption is fragile in this case, as regions with high MMI already had higher levels of investment, employment, and output prior to the earthquake. Equation (5) is designed to test this assumption:

$$y_{jrt} = \sum_{k \neq 2009} \beta_k (\mathbf{1}[t = k] \times \text{MMI}_r) + \gamma_{jr} + \delta_{jt} + \varepsilon_{jrt}. \quad (5)$$

To reduce the influence of these pre-existing trends, I estimate each region’s trend over the pre-earthquake period (2003–2009), project it forward to construct a counterfactual path, and subtract this counterfactual from the actual data. Then I use the resulting residuals to estimate Equation (5). I also use the 2014 FUT tax reform as a placebo shock for a falsification test.

4.3 Inference with few clusters

Because there are only 15 regions, cluster-robust standard errors are downward biased and the asymptotic t -test over-rejects. For the headline coefficients I therefore also report restricted wild cluster bootstrap p -values (Rademacher weights, $B = 999$, the null imposed), following [Cameron et al. \(2008\)](#).

5 Results

5.1 The reconstruction-investment pulse

Table 2 reports the baseline coefficient from (4) for each outcome. Panel A, without region trends, shows large and significant effects on investment, gross output, and employment. Panel B adds region-specific linear trends: the investment coefficient falls from 0.52 to 0.17

and loses significance, and the output and employment effects disappear. Most of Panel A is the differential secular growth of the central regions, not the earthquake, and this result is exactly the concern raised by the long pre-period.

	Investment	Employment	Wage	Establishments	Gross output
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. No region trends</i>					
Post _{<i>t</i>} × MMI _{<i>r</i>}	0.517*** (0.135)	0.165** (0.079)	0.247 (0.156)	0.075 (0.046)	0.279** (0.115)
Observations	414				
<i>Panel B. + Region linear trends</i>					
Post _{<i>t</i>} × MMI _{<i>r</i>}	0.168 (0.102)	0.050 (0.039)	0.171 (0.219)	0.030 (0.028)	-0.041 (0.093)
Observations	414				
<i>Panel C. + Region quadratic trends</i>					
Post _{<i>t</i>} × MMI _{<i>r</i>}	0.205** (0.104)	0.048 (0.042)	-0.290 (0.321)	0.031 (0.027)	-0.043 (0.096)
Observations	414				
Industry × region FE	✓	✓	✓	✓	✓
Industry × year FE	✓	✓	✓	✓	✓

Notes. Cells are 2-digit ISIC industry $\times$ region $\times$ year, sample restricted to Food (10), Wood (16) and Basic metals (24). MMI_r is the region’s area-weighted Modified Mercalli Intensity from the 2010 Maule earthquake (USGS ShakeMap); $\text{Post}_t = \mathbf{1}[t \geq 2010]$. Each entry is a separate WLS regression weighted by cell effective N ; standard errors clustered by region (15 clusters) in parentheses. Panels B and C add full-sample region-specific linear and quadratic time trends; the quadratic panel is a conventional nonlinear-trend robustness, while the principled sensitivity to parallel-trends violations is the HonestDiD analysis in Section 8. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

However, for a transient phenomenon, a single post-earthquake coefficient is not an appropriate measure, as it reduces the results to an average, preventing me from seeing how the results change over time. Therefore, I conducted an event study organized by year. Table 3 and Figure 2 present the results of the event study. Even during the long pre-earthquake period, the investment gap in high-MMI regions remained positive relative to the base year; therefore, the event study remains inconclusive and cannot be shown to strictly follow a parallel pre-event trend. However, it shows that during the reconstruction period, the divergence in investment gaps between severely and lightly affected areas became very large: the investment gap between high and low MMI areas rose to 0.58 in 2014 ($p < 0.01$) and to 0.60 in 2015 ($p < 0.05$), before falling back to near zero in 2016–2017.

These changes track the institutional structure of the reconstruction. Chilean authorities adopted a four-year reconstruction plan, financed by temporarily raising corporate tax rates,

increasing mining royalties, drawing on the sovereign wealth fund, and issuing new bonds (Government of Chile, 2010). The Ministry of Housing and Urban Development (MINVU) committed to rebuilding or repairing 220,000 housing units over four years, but noted that the affected population and housing stock were geographically dispersed across rural areas and that land-ownership and technical constraints made rapid execution difficult. The immediate post-disaster priorities were therefore relief, assessment, subsidies, and approvals; manufacturing establishments recorded the acquisition and replacement of structures, machinery, vehicles, and software only once projects reached the implementation phase. The 2014–2015 investment peak thus reflects the execution of reconstruction rather than an immediate response to the physical shock.

The magnitude of the investment effect in 2014 depends on the method of trend removal: the effect is 0.71 with no detrending, 0.40 with detrending applied to the entire sample, and I tend to use the figure of 0.58, which lies between the two and passed the bootstrap test for wild clustering ($p = 0.027$).

Table 4 shows that the 2014 coefficient is robust across alternative specifications—adding industry $\times$ region trends, restricting to a tight 2007–2017 window, dropping the Santiago Metropolitan Region, and using the full manufacturing sample, where it ranges from 0.56 to 0.88 and remains significant. Under a binary high-intensity ($\text{MMI} \geq 6$) treatment it is larger still (1.87) but also significant. This supports H1.

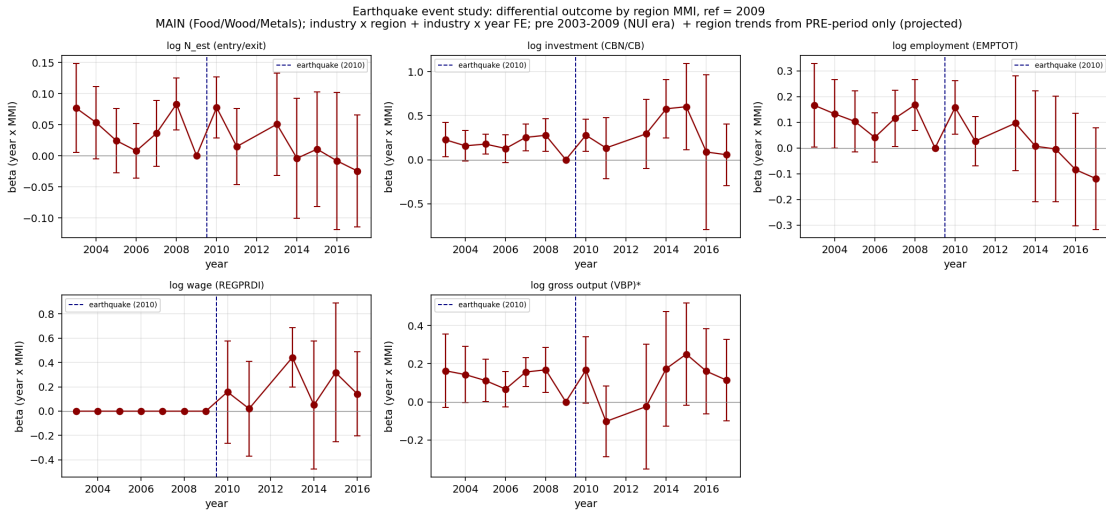


Figure 2: Investment event study with region trends estimated on the pre-period (2003–2009) only and projected forward. The 2014–2015 spike is substantially larger than the remaining pre-period gaps and reverts thereafter.

Table 3: Investment event study: β_k on $\mathbf{1}[t = k] \times \text{MMI}_r$ (reference year 2009)

Year k	$\hat{\beta}_k$	(s.e.)
2003	0.226**	(0.099)
2004	0.158*	(0.088)
2005	0.178***	(0.057)
2006	0.127	(0.081)
2007	0.254***	(0.077)
2008	0.278***	(0.095)
2009	— reference —	
2010 (<i>earthquake</i>)	0.279***	(0.093)
2011	0.134	(0.177)
2013	0.294	(0.199)
2014	0.579***	(0.170)
2015	0.602**	(0.250)
2016	0.088	(0.449)
2017	0.057	(0.178)

Notes. MAIN sample (Food/Wood/Metals). Region linear trends are estimated on the pre-earthquake years (2003–2009) only and projected forward, so the post-quake response does not contaminate the trend. Industry $\times$ region and industry $\times$ year fixed effects included; SE clustered by region. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Robustness of the 2014 reconstruction-investment pulse

Specification	2014 $\hat{\beta}$	2015 $\hat{\beta}$	N
baseline (MAIN, region trend)	0.579*** (0.170)	0.602**	401
R1 industry $\times$ region trends	0.592*** (0.161)	0.579**	401
R2 tight window 2007-2017	0.876*** (0.305)	0.862**	265
R3 drop Santiago (reg 13)	0.582** (0.281)	0.475*	354
R4 binary $\text{MMI} \geq 6$	1.873*** (0.405)	2.069***	401
R5 all manufacturing	0.557*** (0.085)	0.166	2112

Notes. Each row re-estimates the 2014 (and 2015) coefficient on $\mathbf{1}[t = k] \times \text{MMI}_r$ for log investment under one design change, using pre-period-projected detrending. SE (2014) clustered by region in parentheses. The 2014 pulse survives a restricted wild cluster bootstrap (Rademacher, $B = 999$): $p = 0.027$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Reconstruction without reallocation

Table 5 addresses the question posed by H2: Where does the increase in investment manifest itself? The answer is: nowhere. I apply the same pre-period-projected detrending and regression to output, employment, labor productivity, the labor share of output, and the corporate-income-tax profit proxy. The results are all statistically indistinguishable from zero. In other words, reconstruction restored the destroyed capital but did not expand production, alter the level of employment or labor productivity, or change the distribution of income between labor and capital. I interpret this as “replacement”: establishments rebuilt the capital destroyed by the earthquake, restoring it to its pre-earthquake level rather than expanding beyond the original level. This explains why the real and distributional margins remained unchanged, and why (consistent with [Díaz et al., 2024](#)) regional value added consistently fell short of the counterfactual path. The only exception is a slight increase in the wage bill (+0.14, $p < 0.05$). I analyze the possible mechanisms behind this wage increase in Sections 6 and 7. These include positive spillover effects from the local construction industry and the out-migration of unskilled workers.

Table 5: Where the reconstruction shock shows up: incidence decomposition

Outcome (cell, logs)	Post $\times$ MMI	(s.e.)
Investment	0.268*	(0.138)
Gross output (VBP)	0.107	(0.081)
Employment	0.009	(0.067)
Labor productivity (VBP/worker)	0.098	(0.071)
Wage bill (REMPAG)	0.136**	(0.056)
Labor share of output (REMPAG/VBP)	0.029	(0.059)
Corporate income tax (RENIMP, profit proxy)	0.040	(0.141)

Notes. Each row is a separate regression of the avg. post-2010 deviation from the region’s pre-period-projected trend on Post $\times$ MMI, MAIN sample, SE clustered by region. Only investment (capital formation) responds; output, employment, labor productivity, the labor share of output, and the profit proxy do not—the rebuilding restored capital without changing the scale or factor distribution of production. The wage bill drifts up but is fragile (see Section 8); the labor share is flat. Value-added-based labor shares are infeasible (value added is recorded only in 2003–07 and 2014–17); RENIMP is a noisy profit proxy (it nets investment-linked depreciation and the 2014 rate change, the latter absorbed by industry $\times$ year FE).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 The skill premium

Based on the capital-skill complementarity principle, one might expect that the reconstruction of infrastructure following an earthquake would be accompanied by a rise in the skill

premium. However, Table 6 does not support this prediction: the premium effect is negative, averaging -0.19 log points per unit of MMI. The overall wage increase shown in Table 5 stems primarily from a rise in unskilled wages, averaging $+0.26$ and peaking at $+0.52$ in 2014, while skilled wages remained stable. This approach has a limitation: it remains a reduced-form regression that captures the combined impact of equipment, structures, labor, and output on the skill premium in the wake of the earthquake, rather than the independent effect of any single factor.

Table 6: The labor margin: reconstruction raises low-skill wages

	Skilled wage (1)	Unskilled wage (2)	Skill premium (3)=(1)−(2)
Avg. post-2010 $\times$ MMI	0.059	0.258**	-0.199
<i>By sub-period ($\times$ MMI):</i>			
2010–2014	-0.001	0.217**	-0.218
2015–2016	0.165	0.332	-0.167

Notes. Skilled wage = REGTESP (technicians/specialists), unskilled = REGNOCALD (no calificado), 2003–2016. Each coefficient is the avg. post-2010 deviation of the cell-mean log wage from its pre-period-projected region trend, per unit MMI; MAIN sample; SE clustered by region. The skill premium falls because the *unskilled* wage rises (peaking in 2014, Fig. 4), not because the skilled wage falls — the canonical capital–skill complementarity prediction (a rising premium) is rejected. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Replacement, and the Missing Wage Mechanism

This section primarily clarifies two points: First, the investment pulse following an earthquake represents a replacement of destroyed capital rather than further growth from the original level. Second, my analysis using household data reveals that the conclusion regarding an upward shift in unskilled wages is fragile. Taken together, these two points indicate that investment is linked to the demand for post-disaster reconstruction, rather than to a capital-driven shift in labor demand.

Investment represents a replacement. Table 7 examines the asset classes to which the increase in investment in 2014 was allocated: structures accounted for the largest share ($+0.82$), while machinery ($+0.49$) and broad equipment ($+0.52$) also saw significant increases (see Figure 3). Panel B shows that the equipment coefficient was 0.34 higher than its pre-period trend during both the 2010–2014 and 2015–2017 periods, while the structures coefficient was lower in the latter period. Therefore, this table serves as key evidence supporting the paper’s argument: reconstruction covered structures and equipment, but did

not affect the distribution of output, employment, or skills, primarily because reconstruction merely restored the capital base rather than expanding or developing it beyond its original level.

<u>Table 7: Mechanism: reconstruction is low-skill-labor intensive</u>		
<i>Panel A. 2014 investment pulse by asset class ($\times$ MMI)</i>		
	$\hat{\beta}$	(s.e.)
Structures	0.820***	(0.225)
Machinery	0.488***	(0.129)
Equipment	0.525***	(0.133)
<i>Panel B. Net accumulation: avg. post-2010 deviation ($\times$ MMI)</i>		
	2010–14	2015–17
Total investment	0.299***	0.232
Structures	0.292*	0.123
Equipment	0.333***	0.352

Notes. From the pre-period-projected detrended residual, MAIN sample, SE clustered by region. Panel A: the 2014 pulse spans both structures and equipment. Panel B: equipment accumulation persists (net stock rises), while structures is transient. So the absence of a skill response (Table 6) is not because no equipment was bought. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

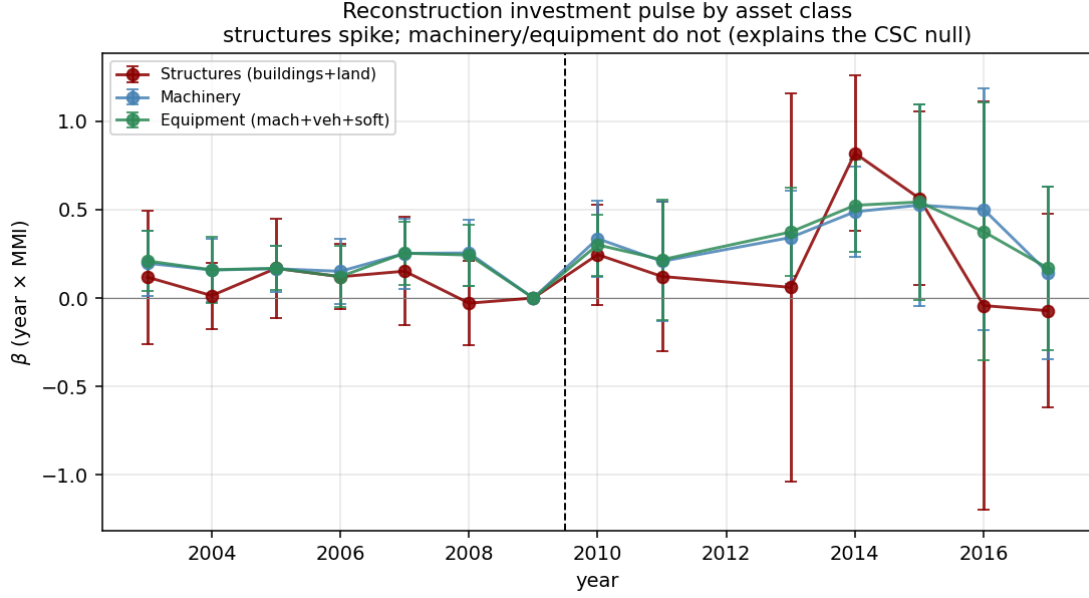


Figure 3: Reconstruction investment pulse by asset class. Both structures and equipment rise significantly; the null on skill is therefore not due to an absence of equipment investment.

The rise in unskilled wages did not originate in the construction industry.

As shown in Figure 4, the only labor-side variable that changed was unskilled wages, which gradually increased and peaked alongside investment in 2014, while skilled wages remained stable. One plausible mechanism is the spillover effect from the local labor market: following the disaster, there was a greater need for unskilled labor to rebuild infrastructure and factories. Consequently, the construction boom in the hardest-hit areas would have stimulated wages for local unskilled workers, forcing manufacturing plants to follow suit with wage increases. I directly tested this hypothesis using household survey data and found that, in the more severely affected regions, post-earthquake employment in the construction sector did not increase (+0.02, $p = 0.36$), nor did construction wages rise. On the contrary, they declined (-0.04 , $p = 0.09$). The event study also revealed no break at the time of the earthquake. At the same time, I found that the rise in unskilled wages stemmed primarily from the food industry, rather than the wood products or basic metals industries, the two sectors closely related to construction. The conclusions are presented in Table 8.

Table 8: Robustness of the wage mechanism: alternative explanations

	Avg. post $\times$ MMI	2014 $\times$ MMI	N
<i>Panel A. Composition and placebo checks</i>			
Log establishments	—	-0.018 (0.052)	—
Log employment	—	-0.019 (0.114)	—
Log gross output	—	0.153 (0.151)	—
Fake 2014 shock within post-period	—	0.233 (0.250)	141
<i>Panel B. Wage decomposition and internal placebo</i>			
Unskilled wage	0.262** (0.113)	0.515*** (0.149)	339
Skilled wage	0.073 (0.105)	0.026 (0.196)	347
Skill-premium effect	-0.188* (0.097)	-0.528** (0.252)	331
Unskilled wage + region-year controls	0.147 (0.130)	0.388** (0.154)	339
<i>Panel C. Industry heterogeneity</i>			
Food only	0.287** (0.137)	0.567*** (0.160)	164
Wood/basic metals	0.109 (0.267)	0.095 (0.572)	175

Notes. Coefficients are per unit of regional MMI after pre-period-projected detrending. The wage rows use cell-mean log wages in the MAIN sample unless otherwise noted. Panel A reports 2014 event-study coefficients for non-wage outcomes and a fake-shock placebo estimated only within the post-earthquake period. Region-year controls are leave-one-industry-out ENIA manufacturing aggregates: log establishments, employment, output, investment, and regional production wage. Standard errors, clustered by region, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

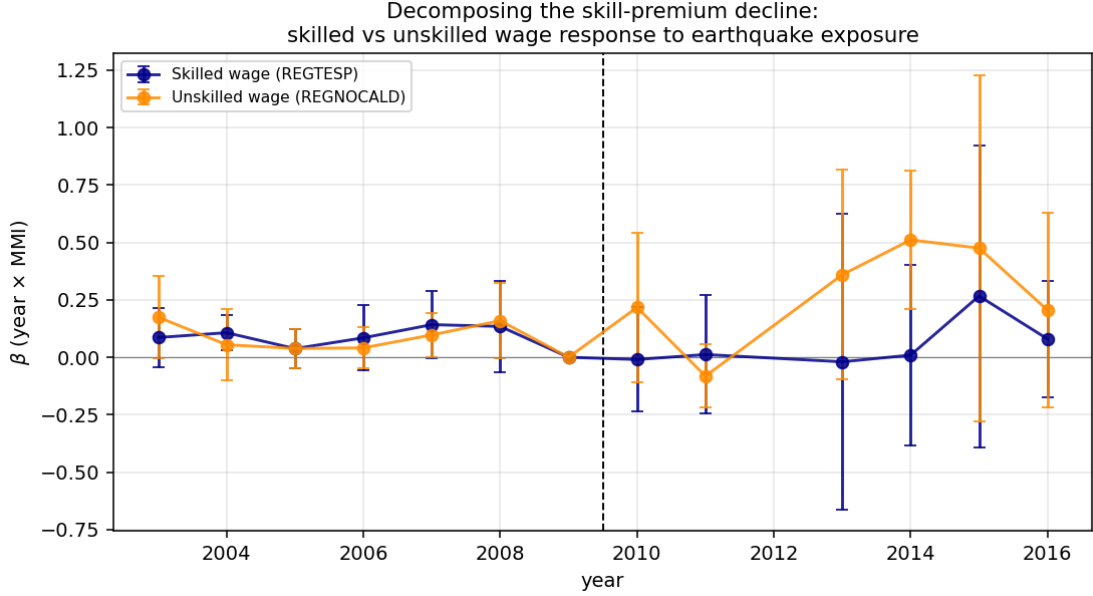


Figure 4: Decomposing the skill-premium decline: the unskilled wage rises and peaks in 2014, in lockstep with the investment pulse, while the skilled wage is flat.

There are three potential limitations to note. First, the classification criteria for skilled and unskilled workers in the original data are relatively crude. Second, the skill premium reflects both labor demand and labor supply: when demand for unskilled labor rises, wages for unskilled workers increase, and, all else being equal, the skill premium declines. When the supply of unskilled labor increases (i.e., if more unskilled workers enter the disaster-affected area), wages for unskilled workers fall, and the skill premium rises. Conversely, if the earthquake causes workers to become displaced and leave the severely affected areas, the net effect on the skill premium is ambiguous. The key lies in the post-earthquake mobility of these two categories of workers—specifically, whether they left the severely affected areas. I discuss this point in Section 7. Third, this paper draws inferences based solely on data from 15 regions, and this is the primary reason I employed the bootstrapping method.

7 The Out-Migration Channel

There is another possible mechanism to explain why unskilled wages rise: the out-migration effect. If unskilled workers leave the hardest-hit areas, the local supply of unskilled labor contracts, causing unskilled wages to rise. However, this mechanism is unrelated to reconstruction. The out-migration effect and the reconstruction mechanism mentioned earlier appear to have the same directional impact on wages, while their difference lies in the num-

ber of workers: if the rise in unskilled wages is primarily due to reconstruction, the number of unskilled workers will increase. If it is caused by the out-migration effect, the number will decrease.

ENIA cannot address this issue because it stopped collecting statistics broken down by skill level after 2007. Therefore, I use another dataset here: the National Socioeconomic Characterization Survey (CASEN). CASEN records region, educational attainment, and population expansion weights, allowing me to calculate the stock of skilled and unskilled workers in each region before and after the earthquake. CASEN survey data is only available for 2003, 2006, 2009, 2011, 2013, 2015, and 2017, with a lower frequency of data collection than ENIA. Therefore, I primarily use it for robustness checks. I adopt the same specifications as in the main analysis ($\text{Post} = \mathbf{1}[t \geq 2010]$, where the 2009 wave is classified as pre-earthquake) and divide regions into the 13 major regions as defined prior to 2007. Skill levels are still categorized by educational attainment: unskilled workers have a basic education or lower, while skilled workers have a higher education background.

Table 9 and Figure 5 present the results, which support the demand-side interpretation. The working-age unskilled population in severely affected areas did not decline after the earthquake: the $\text{Post} \times \text{MMI}$ coefficient was $+0.002$, which is statistically indistinguishable from zero. Between 2003 and 2017, the unskilled population in areas with high MMI maintained a small positive difference, and no breakpoint occurred at the time of the shock. The coefficient for 2006 in the event study was positive, but the combined test of the pre-event trend (i.e., that the coefficients for both 2003 and 2006 were equal to zero) was not rejected ($F = 2.60$, $p = 0.12$). To rule out the out-migration confound, I used the same HonestDiD test as in the main results. The results show that when $\bar{M} = 1$, the robust confidence interval for the relative magnitude of the average post-earthquake effect is $[-0.07, 0.12]$. Therefore, even if I allow for a parallel-trend violation after the earthquake that is as large as the worst-case scenario before the earthquake, the unskilled population in the more severely affected areas would decline by no more than approximately 7 percent per unit of MMI. This figure is far insufficient to account for the level of unskilled wage increases mentioned in Section 6. At the same time, the data in Table 9 indicate that the unskilled labor force actually increased slightly after the earthquake, and its share of the total population remained unchanged. Therefore, an out-migration effect can be ruled out. Furthermore, the slight decline in the total population (-0.009 per unit of MMI) is attributable not to the unskilled labor force but to the skilled labor force. At the same time, wages for the skilled labor force did not change significantly despite the decline in the size of this group. The mechanisms underlying this phenomenon are beyond the scope of this paper and may thus serve as a topic for future discussion and research. This conclusion still has some data-related limitations: the

CASEN survey is conducted infrequently, and the small number of clusters (13 regions in the harmonized CASEN scheme) may limit the precision of the estimates.

Table 9: Out-migration check: regional population by skill (CASEN, 2003–2017)

	Unskilled pop.	Skilled pop.	Total pop.	Unsk. share
	(1)	(2)	(3)	(4)
$\text{Post}_t \times \text{MMI}_r$	0.002 (0.006)	-0.009 (0.010)	-0.009*** (0.003)	-0.001 (0.003)
Joint pre-trend test, unsk.	$F = 2.60$ ($p = 0.12$)			
HonestDiD robust CI ($\bar{M} = 1$)	[-0.07, 0.12]			
Region $\times$ year cells	91			

Notes. Cells are region $\times$ CASEN-wave; outcomes are the log of the population-weighted count of working-age (15–64) persons in each group, except column (4) (the unskilled share). Skill is defined by education: unskilled = basic schooling or less, skilled = tertiary. $\text{Post}_t = \mathbf{1}[t \geq 2010]$ (the 2009 wave is pre-earthquake; the first post-earthquake wave is 2011). Regions are harmonized to the pre-2007 13-region definition; SE clustered by region. A negative coefficient on column (1) would indicate unskilled out-migration from harder-hit regions; the coefficient is instead near zero. The joint pre-trend test (the 2003 and 2006 event-study coefficients equal zero) is not rejected, and the HonestDiD relative-magnitudes robust CI for the average post-earthquake effect at $\bar{M} = 1$ (Rambachan & Roth, 2023) still rules out any large unskilled out-migration. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

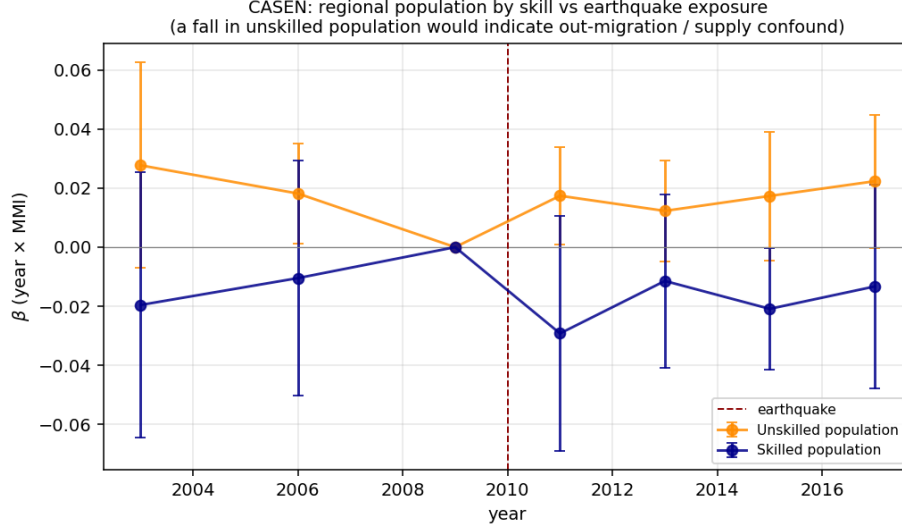


Figure 5: CASEN: regional population by skill versus earthquake exposure. A fall in the unskilled population in harder-hit regions would indicate out-migration; the coefficient is instead near zero.

8 Sensitivity to Parallel Trends

The identification in this paper relies on the parallel-trend assumption, namely, that after controlling for region-specific trends, the growth trends of various variables (investment, output, employment, wages/skill premiums, etc.) in regions with high MMI and low MMI are parallel. If the pre-shock coefficients in the regression model are approximately equal to zero and statistically insignificant, then the pre-shock parallel-trend assumption holds. However, I found that not all pre-earthquake coefficients were equal to zero. To address this issue, I employed the sensitivity analysis method proposed by [Rambachan & Roth \(2023\)](#) to assess the core empirical conclusion of this paper, namely, to what extent the investment spike in 2014, resulting from post-earthquake reconstruction, depends on the assumption of prevailing trends. This method moves beyond the original binary classification of “parallel” versus “non-parallel” to determine the extent to which the pre-earthquake trend can deviate from parallelism while the 2014 effect remains statistically significant. I applied this approach to the raw event-study estimates, summarizing the results in Table 10 and Figure 6.

There are two natural ways to impose constraints here. The smoothness constraint $\Delta^{SD}(M)$ limits the change in the slope of the divergence trend in each period to no more than M , meaning that the trend line is not allowed to turn sharply and can only change gradually. When $M = 0$, the trend is forced to be a straight line, which is entirely consistent with the linear detrending process I described earlier. The relative magnitude constraint

$\Delta^{RM}(\bar{M})$, on the other hand, uses the actual deviations observed before the earthquake as a baseline and limits the magnitude of deviations from the post-earthquake parallel trend to within $\bar{M}$ times the maximum pre-earthquake deviation. The advantage of this approach is that it is not affected by the uneven distribution of years caused by the exclusion of the 2012 wave. These two constraints operate from the perspectives of “how sharply the trend can turn” and “how large the post-earthquake deviation is relative to the pre-earthquake value,” respectively. I apply both simultaneously to ensure robustness.

Under the linear-trend benchmark $\Delta^{SD}(0)$, the robust confidence sets for the investment coefficient and the unskilled wage coefficient in 2014 both exclude zero ($[0.28, 0.75]$ and $[0.19, 0.80]$, respectively), consistent with the estimates preferred in this paper. However, these conclusions are fragile in the face of larger deviations: the breakdown values for both results are approximately $\bar{M} \approx 0.5$, which is lower than the commonly used benchmark of $\bar{M} = 1$ (under which post-earthquake violations as large as the most severe pre-earthquake violations are permitted, at which point both confidence sets include zero).

I also used another, more refined estimation method to assess robustness. Since locally smoothed trends vary approximately linearly between adjacent years, the difference $\beta_{2014} - \frac{1}{2}(\beta_{2013} + \beta_{2015})$ can account for such a trend, thereby isolating any “spikes” in a single year. However, this does not help in this case (Table 10, last row): the difference itself is not significant because the effect is a multi-year upward shift between 2013 and 2015, rather than a spike in a single year; thus, subtracting 2014 from its adjacent years removes the signal along with the trend. Combined with standard linear and quadratic trend specifications (Table 2, Panels B–C, under which the investment effect barely survives), I believe the data support the following conclusion: as long as deviations from the parallel trend are mild and smooth, the findings regarding the reconstruction of the investment pulse and the constancy of unskilled wages are solid. However, once deviations become too large, these findings become fragile.

Table 10: HonestDiD sensitivity of the 2014 effect to parallel-trends violations

	Investment	Unskilled wage
Original CI (PT imposed)	[0.37, 1.05]	[0.20, 0.83]
<i>Relative magnitudes $\Delta^{RM}(\bar{M})$:</i>		
$\bar{M} = 0.5$	[-0.15, 1.53]	[-0.18, 1.17]
$\bar{M} = 1.0$	[-0.81, 2.18]	[-0.72, 1.69]
breakdown $\bar{M}$	0.50	0.50
<i>Smoothness $\Delta^{SD}(M)$:</i>		
$M = 0$ (linear trend)	[0.28, 0.75]	[0.19, 0.80]
$M = 0.05$	[-0.70, 1.68]	[-0.11, 1.73]
breakdown M	0.02	0.05
Spike contrast, original CI	[-0.13, 0.46]	[-0.21, 0.40]

Notes. Robust 95% confidence sets for the 2014 (year $\times$ MMI) coefficient from the raw event study, following [Rambachan & Roth \(2023\)](#). $\Delta^{RM}(\bar{M})$ bounds the post-period violation of parallel trends by $\bar{M}$ times the largest pre-period violation; $\Delta^{SD}(M)$ bounds the change in the differential trend’s slope by M per period ($M = 0$ is exactly linear, equivalent to the paper’s pre-period detrending). Breakdown is the value at which the robust CI first includes zero. **Bold** intervals exclude zero. The “spike contrast” targets $\beta_{2014} - \frac{1}{2}(\beta_{2013} + \beta_{2015})$, which nets out a locally smooth trend but also the multi-year pulse, and is uninformative here.

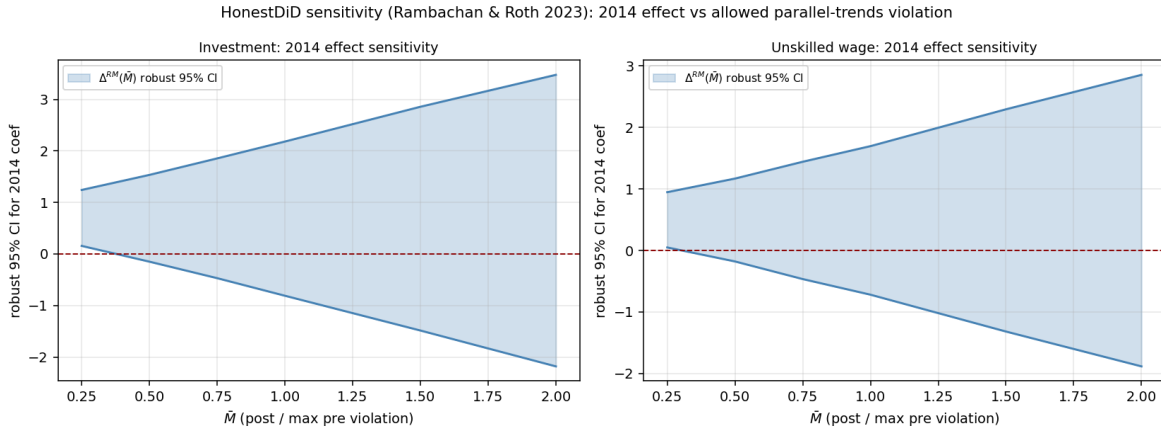


Figure 6: HonestDiD sensitivity ([Rambachan & Roth, 2023](#)) of the 2014 coefficients to relative-magnitude violations of parallel trends. The robust 95% confidence set widens with $\bar{M}$; it excludes zero only for $\bar{M}$ below about 0.5.

9 Discussion

Post-earthquake reconstruction focused primarily on replacing the damaged capital rather than rebuilding to the original level of development. Investment increased to rebuild and replace capital, but there were no changes in output, employment, the allocation of key factors, or wages in any robust sense. Under these circumstances, the marginal returns to capital and labor returned to pre-earthquake levels, so neither the prices nor the quantities of these two factors changed significantly. At the same time, I found that when simulating a counterfactual path in which the earthquake shock did not occur, the actual level of development remains below that of the counterfactual path. This is consistent with the findings of [Díaz et al. \(2024\)](#), namely that investment offsets the damage rather than building upon the original foundation. Furthermore, because investment is not aimed at development, its impact on the labor force and income distribution is minimal.

Alternative explanations. Theoretically, several other mechanisms could also explain the findings of this paper. (a) *Nominal and real wages.* The data used in this paper are based on nominal wages, so inflation during the reconstruction period may have led me to overestimate changes in unskilled or skilled wages. However, the skill premium differences out common inflation, so the skill-premium conclusion is not affected by inflation. (b) *Insurance and reconstruction transfer payments.* Public reconstruction spending and insurance payouts boosted consumption and purchasing power in severely affected areas, leading to increased demand for local goods and services. This may account for part of the change in local demand, but it does not invalidate my regression results. (c) *Survivorship bias.* If larger establishments were more likely to survive the earthquake, while smaller ones were more likely to disappear and thus not be included in post-earthquake statistics, then the post-earthquake data would be skewed toward surviving establishments. However, [Table 2](#) shows that the change in the number of establishments is zero and not statistically significant. (d) *Output demand for construction materials.* Reconstruction increases the construction industry’s demand for building materials such as lumber, cement, and metals. However, wage increases were concentrated in the food industry, the sector least associated with reconstruction. The specific reason for wage increases in the food industry is not the focus of this paper but may serve as a new research idea for the future.

Soil-correlated agricultural shocks. A more subtle concern is that shaking intensity is correlated with soft alluvial soil, and the alluvial and volcanic soils of Chile’s Central Valley are also highly fertile ([Casanova et al., 2013](#)), so a regional agricultural boom around 2014 could have raised demand for food-related manufacturing and mimicked the reconstruction pulse. Because such a shock varies at the region-by-year level, it is not absorbed by the

industry $\times$ year fixed effects and it survives linear detrending, so I test it directly. In CASEN, neither the agricultural employment share nor the agricultural wage responds to Post $\times$ MMI (+0.003, $p = 0.41$, and +0.014, $p = 0.58$), and agricultural employment, although it rises, drifts up monotonically through 2017 rather than peaking in 2014. In ENIA, food-sector investment reproduces the aggregate pulse, rising to +0.67 in 2014 and +0.71 in 2015 before reverting, whereas food-sector output shows no significant 2014 break (+0.18, $p = 0.30$). Since a genuine demand-driven agricultural boom would raise farm wages and food output and would share the pulse’s peak-and-revert timing, the data do not support this channel.

2014 tax reform. In 2014, the Chilean government enacted a new corporate tax policy (Law 20,780), which could have led to front-loading—that is, companies rushing to make investments before the tax reform officially took effect, resulting in a peak in investment in 2014. Moreover, such front-loading was more likely to occur in disaster-stricken areas where businesses were already more concentrated. However, the paper has ruled out this possibility on two grounds. First, the assets that establishments would front-load are those that can be purchased in advance, such as machinery. Therefore, if a 2014 investment peak were indeed caused by anticipatory front-loading, that peak should have been dominated by equipment. However, the findings in Table 7 show that the investment peak was dominated by structures, which establishments cannot purchase in advance. Second, no further spikes should have occurred after the tax reform was implemented, so the investment peak should have subsided in 2014. However, my data show that the investment peak persisted into 2015. Therefore, I believe that the 2014 tax reform was not the primary cause of the investment peak and does not affect the findings of this paper.

Spillover effects. In this paper, the severely affected areas with high MMI are treated as the treatment group, while the lightly affected areas with low MMI are treated as the control group. However, the lightly affected areas were not entirely unaffected. Post-earthquake reconstruction requires large quantities of cement, lumber, steel, and workers, all of which are likely to be transferred from the lightly affected areas to the severely affected areas. In other words, because the lightly affected areas are supplying materials to the severely affected areas, manufacturing demand rises, pulling up output and investment, which narrows the gap between the treatment group and the control group, and causes β to underestimate the true effect. At the same time, the transfer of labor from the lightly affected areas to the severely affected areas reduces employment in the former, widening the gap between the control group and the treatment group, and that causes β to overestimate the true effect.

I test for the spillover directly with a spatial-lag-of-treatment (SLX) specification, adding a second treatment term, Post $\times$ MMI for a region’s immediate geographic neighbors, to the incidence regressions. Table 11 reports the result, and the result is that the spillover cannot

be signed with this design. Because Chile is long from north to south but very narrow from east to west, its geography is almost linear, with the various regions essentially lined up end-to-end. Therefore, a region’s own shaking and its neighbors’ shaking are nearly collinear ($r \approx 0.98$), so the own- and neighbor-exposure terms are not separately identified. The investment neighbor coefficient is 0.51 with a standard error above one, $p = 0.64$, and the output own-coefficient inflates from a precise null of 0.11 to 1.06 while a mechanically “significant” neighbor term of -0.97 appears with the opposite sign. This means the standard errors explode and the augmented own-coefficients become unstable. Therefore, the sign of the spillover is ambiguous.

Table 11: Spatial-lag (SLX) spillover test: own versus neighbor exposure

Outcome (cell, logs)	Own Post $\times$ MMI		Neighbor Post $\times$ MMI
	Baseline	+ Neighbor	(augmented)
Investment	0.268* (0.138)	-0.228 (1.036)	0.507 (1.091)
Gross output (VBP)	0.107 (0.081)	1.060*** (0.367)	-0.975** (0.390)
Employment	0.009 (0.067)	0.236 (0.470)	-0.232 (0.498)

Notes. “Neighbor exposure” is the mean MMI of a region’s immediate north-south geographic neighbors (Chile is effectively one-dimensional). Each outcome is regressed, on the pre-period-projected-detrended footing of Table 5 (MAIN sample, SE clustered by region), first on own Post $\times$ MMI alone (“Baseline”), then on own and neighbor Post $\times$ MMI jointly (“+ Neighbor”/“augmented”). Own and neighbor MMI are collinear across regions ($r \approx 0.98$), so the two terms are not separately identified: the augmented own-coefficients are unstable (the output own-coefficient inflates with an opposite-signed neighbor term) and standard errors are large (the investment neighbor coefficient has a standard error above one). The sign of any cross-region spillover is therefore not resolvable with this design. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Limitations. (i) Since 2008, the establishment identifiers in the ENIA data have changed annually, making it impossible for me to identify the same establishment across different years. Consequently, I am unable to construct an establishment panel and can only aggregate the data into cells organized by “industry $\times$ region $\times$ year.” Therefore, based on the available data, I cannot distinguish whether the effects I identify stem from internal adjustments within a specific establishment or from overall changes in the industry average. (ii) ENIA applies multiplicative confidentiality noise, which increases the variance. To address this, I performed a bootstrap analysis. (iii) Wages for unskilled workers rose during the post-earthquake reconstruction period, while wages for skilled workers remained largely unchanged. The specific mechanism behind this phenomenon has not yet been elucidated.

Only two mechanisms have been ruled out: the local construction industry development mechanism mentioned in Section 6 and the out-migration effect of unskilled labor mentioned in Section 7. Further possible mechanisms remain to be explored in future research. (iv) Regardless of the type of publicly available data, none of it covers the working hours of individual workers. Therefore, the analysis of the labor force in this paper is limited to the single dimension of “number of workers.”

10 Conclusion

I used the 2010 Maule earthquake in Chile as an exogenous shock to examine its impact on Chilean manufacturing and how the sector adjusted during post-disaster reconstruction. The results show that regions most severely affected by the disaster experienced a temporary spike in investment, peaking in 2014. At the same time, there were no significant changes in output, employment, labor productivity, the labor share of output, or the profit proxy. The rise in wages for unskilled workers was the only labor market change observed, but the mechanism behind it remains unclear. Furthermore, this paper points out that post-earthquake reconstruction merely restored the capital destroyed by the earthquake, rather than building upon the existing foundation. Therefore, the post-earthquake investment peak represents “reconstruction” rather than “labor reallocation.”

References

- Aksoy, C. G., Chupilkin, M., Koczan, Z., & Plekhanov, A. (2024). Unearthing the economic and social consequences of earthquakes. *IZA Discussion Paper No. 17108*.
- Bárcena, A., Prado, A., López, L., & Samaniego, J. (2010). *The Chilean Earthquake of 27 February 2010: An Overview*. Santiago: Economic Commission for Latin America and the Caribbean (ECLAC).
- Belasen, A. R., & Polachek, S. W. (2009). How disasters affect local labor markets: The effects of hurricanes in Florida. *Journal of Human Resources*, 44(1), 251–276. [doi:10.3368/jhr.44.1.251](https://doi.org/10.3368/jhr.44.1.251)
- Benguria Garibaldi, S. (2022). Shaky growth: Chile’s earthquake and its effect on GDP. Master’s thesis, Stockholm School of Economics.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427. [doi:10.1162/rest.90.3.414](https://doi.org/10.1162/rest.90.3.414)
- Casanova, M., Salazar, O., Seguel, O., & Luzio, W. (2013). *The Soils of Chile*. World Soils Book Series. Dordrecht: Springer. [doi:10.1007/978-94-007-5949-7](https://doi.org/10.1007/978-94-007-5949-7)
- Díaz, D., Paniagua, P., & Larroulet, C. (2024). Earthquakes and the wealth of nations: The cases of Chile and New Zealand. *Working paper*, arXiv:2405.12041.
- Fuest, C., Peichl, A., & Siegloch, S. (2018). Do higher corporate taxes reduce wages? *American Economic Review*, 108(2), 393–418. [doi:10.1257/aer.20130570](https://doi.org/10.1257/aer.20130570)
- Government of Chile. (2010). *Plan de Reconstrucción Chile Unido Reconstruye Mejor* [A United Chile Rebuilds Better: National Reconstruction Plan]. Santiago: Gobierno de Chile.
- Krusell, P., Ohanian, L. E., Ríos-Rull, J.-V., & Violante, G. L. (2000). Capital-skill complementarity and inequality. *Econometrica*, 68(5), 1029–1053. [doi:10.1111/1468-0262.00150](https://doi.org/10.1111/1468-0262.00150)
- Lewis, E. (2011). Immigration, skill mix, and capital-skill complementarity. *Quarterly Journal of Economics*, 126(2), 1029–1069. [doi:10.1093/qje/qjr011](https://doi.org/10.1093/qje/qjr011)
- Notowidigdo, M. J. (2020). The incidence of local labor demand shocks. *Journal of Labor Economics*, 38(3), 687–725. [doi:10.1086/706048](https://doi.org/10.1086/706048)

- Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555–2591. doi:10.1093/restud/rdad018
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3), 305–322. doi:10.1257/aeri.20210236
- Suárez Serrato, J. C., & Zidar, O. (2016). Who benefits from state corporate tax cuts? *American Economic Review*, 106(9), 2582–2624. doi:10.1257/aer.20141702
- U.S. Geological Survey. (2010). *M 8.8 — offshore Bio-Bio, Chile (27 February 2010): Earthquake summary and ShakeMap* (Event ID us2010tfan). Reston, VA: USGS. Retrieved from <https://earthquake.usgs.gov/earthquakes/eventpage/us2010tfan>.
- Wald, D. J., & Allen, T. I. (2007). Topographic slope as a proxy for seismic site conditions and amplification. *Bulletin of the Seismological Society of America*, 97(5), 1379–1395. doi:10.1785/0120060267
- Wald, D. J., Quitoriano, V., Heaton, T. H., Kanamori, H., Scrivner, C. W., & Worden, C. B. (1999). TriNet “ShakeMaps”: Rapid generation of peak ground motion and intensity maps for earthquakes in southern California. *Earthquake Spectra*, 15(3), 537–555. doi:10.1193/1.1586057

A Data construction

A.1 Raw ENIA files and sample window

This paper is based on data from the Annual National Industrial Survey (*Encuesta Nacional Industrial Anual*, ENIA) collected by the Chilean National Institute of Statistics (INE). The raw data are collected annually and are divided into three parts: Part 1 records the establishment identifier and classification information (region, industry, and size); Part 2 records the economic accounts, namely gross output, fixed-asset acquisitions (investment) by asset class, the wage bill, and taxes; and Part 3 records the detailed labor variables, namely total employment and wages by occupational category. The main outcome variables in this paper come from Part 2 (investment, gross output, the wage bill, and the corporate income tax used as a profit proxy); Part 3 provides employment together with the skilled and unskilled wages behind the skill premium; and Part 1 provides the region and industry codes used to define the cells.

I first convert each annual database into a single wide-format establishment-level file. The specific processing steps are: standardizing the establishment identifier, removing duplicate IDs, decoding numeric fields that were exported as byte strings in 2009, and merging the three parts through an outer join on the establishment identifier. The working sample covers the years 2003–2017, excluding 2012 because the establishment response rate to the questionnaire was markedly lower than in other years.

The data used in this paper span two different file regimes. Files prior to 2008 use stable establishment identifiers and follow the ISIC Rev. 3 industry classification. The publicly available FUSION files from 2008 onward use annual identifiers that are not consistent across years; I therefore do not construct an establishment-level panel for the period after 2007. Instead, all regressions use repeated cross-sections composed of “2-digit industry $\times$ region $\times$ year” cells.

A.2 Industry harmonization

The industrial classification standard is not the same across all years. The files I recovered from before 2013 are coded in ISIC Rev. 3 (or Rev. 3.1), while the later files use ISIC Rev. 4. I therefore map the pre-2013 codes to Rev. 4 using the official United Nations concordance table. When one Rev. 3 code corresponds to more than one Rev. 4 code, I split the establishment across those Rev. 4 industries using activity-share weights, so that no establishment is double-counted. All of the analysis is then carried out at the 2-digit Rev. 4 level. As in Section 3, I restrict the main sample to food products (10), wood products (16),

and basic metals (24), because these three industries appear in both high- and low-intensity regions and therefore support the within-industry, across-region comparison. I use the full 2-digit manufacturing sample as a robustness check.

A.3 Cell aggregation and outcome construction

I aggregate the establishment-level data into cells defined by 2-digit industry, region, and year. Let i index establishments in industry j , region r , and year t , and let w_i be the activity-share weight from the industry harmonization. For total outcomes, such as investment, employment, output, and the effective number of establishments, I take the weighted cell sum:

$$Y_{jrt}^{\text{sum}} = \sum_{i \in (j,r,t)} w_i y_i.$$

For wage outcomes, I instead take the weighted mean over establishments with positive, non-missing wages:

$$Y_{jrt}^{\text{mean}} = \frac{\sum_{i \in (j,r,t)} w_i y_i}{\sum_{i \in (j,r,t)} w_i}.$$

I then use the log of the positive cell values as the regression outcomes, except for the skill premium, which I define as the log skilled wage minus the log unskilled wage.

Table 12 lists the main variables and their source fields. The most important construction issue is investment. Through 2014, purchases of new capital goods are reported in the CBN* fields; from the 2015 form redesign onward, the same items are reported under CB* fields. I treat this as a field rename rather than a stop in reporting, and I combine the two series into one. Any national rescaling or form-level break in 2015 is common to an industry-year and is absorbed by the industry $\times$ year fixed effects.

A.4 Earthquake intensity

I measure earthquake exposure with the USGS ShakeMap for the February 27, 2010 Maule earthquake. I overlay the ShakeMap MMI polygons on the GADM 4.1 administrative-region polygons, after reprojecting both layers to a planar equal-area coordinate system so that the areas are computed correctly. Each region's intensity is the area-weighted mean MMI:

$$\text{MMI}_r = \frac{\sum_{g \in r} \text{MMI}_g \cdot \text{area}_{gr}}{\sum_{g \in r} \text{area}_{gr}},$$

where g indexes the ShakeMap polygons that intersect region r . For regions outside the felt area of the earthquake, I assign a floor value of 2.0, which corresponds to weak or essentially

Table 12: Variable construction and source fields

Concept	ENIA source fields	Cell construction
Effective establishments	Harmonized establishment weights	Weighted sum of w_i
Gross investment	CBNTER, CBNEDI, CBNMAQ, CBNVEH, CBNMUE, CBNSOF; from 2015: CBTER, CBEDI, CBMAQ, CBVEH, CBSOF	Weighted cell sum of capital-good purchases
Structures	Land and buildings fields, mainly CBNTER, CBNEDI (old schema) and CBTER, CBEDI (new schema)	Weighted cell sum
Machinery	CBNMAQ and CBMAQ	Weighted cell sum
Equipment	Machinery, vehicles, and software fields	Weighted cell sum
Employment	EMPTOT; for years without EMPTOT, fall-back TOTHOM+TOTMUJ	Weighted cell sum
Gross output	VBP; VBPb/VBPB renamed to VBP where needed	Weighted cell sum
Production wage	REGPRDI	Weighted positive cell mean
Skilled wage	REGTESP (technicians/specialists)	Weighted positive cell mean
Unskilled wage	REGNOCALD (<i>no calificado</i>)	Weighted positive cell mean
Skill premium	REGTESP and REGNOCALD	$\log(\text{skilled}) - \log(\text{unskilled})$

Notes. Field names refer to the harmonized wide ENIA files created from the annual raw databases. The table reports the variables used in the main analysis; additional asset subclasses and robustness variables are built in the same way.

unfelt shaking, so that they serve as low-intensity controls rather than being dropped. To keep the region coding consistent across the ENIA sample, I use the historical fifteen-region structure, so Ñuble is still counted as part of Bío-Bío.

A.5 Detrending, weights, and inference

The main threat to identification is that the hardest-hit central regions were already growing faster before the earthquake. To deal with this, I estimate each region’s linear trend using only the pre-earthquake years (2003, 2004, 2005, 2006, 2007, 2008, and 2009), and I project this trend forward. I then run the event-study and mechanism regressions on the deviations from this pre-period-projected path. I do not estimate the trend on the whole sample, because that would let the post-earthquake reconstruction pulse affect the counterfactual slope and pull the treatment effect toward zero.

I weight the regressions by cell effective size, clipping the cell weight at one for numerical stability, and I cluster the standard errors by region. Because there are only fifteen region clusters, I read the cluster-robust asymptotic standard errors cautiously, and for the headline investment pulse, the 2014 unskilled-wage response, and the equipment-accumulation check I also report restricted wild cluster bootstrap p -values with Rademacher weights.

Finally, ENIA applies multiplicative confidentiality noise to the establishment records. This noise inflates the variance of establishment-level measures, especially investment, but it does not create a systematic difference-in-differences bias in expectation, as long as the perturbation is not correlated with earthquake intensity conditional on the fixed effects. All of the tables and figures in the paper are produced by the replication scripts, which go from the raw annual databases through extraction, harmonization, cell construction, intensity merging, estimation, and table and figure generation.